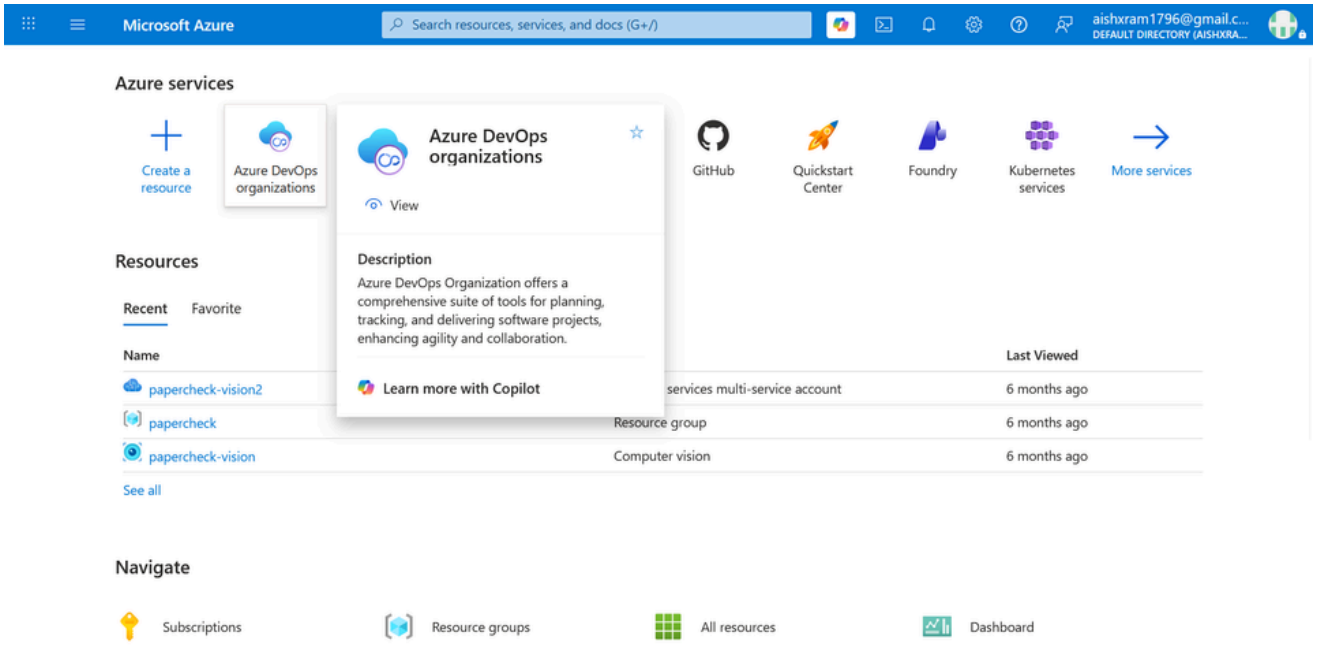
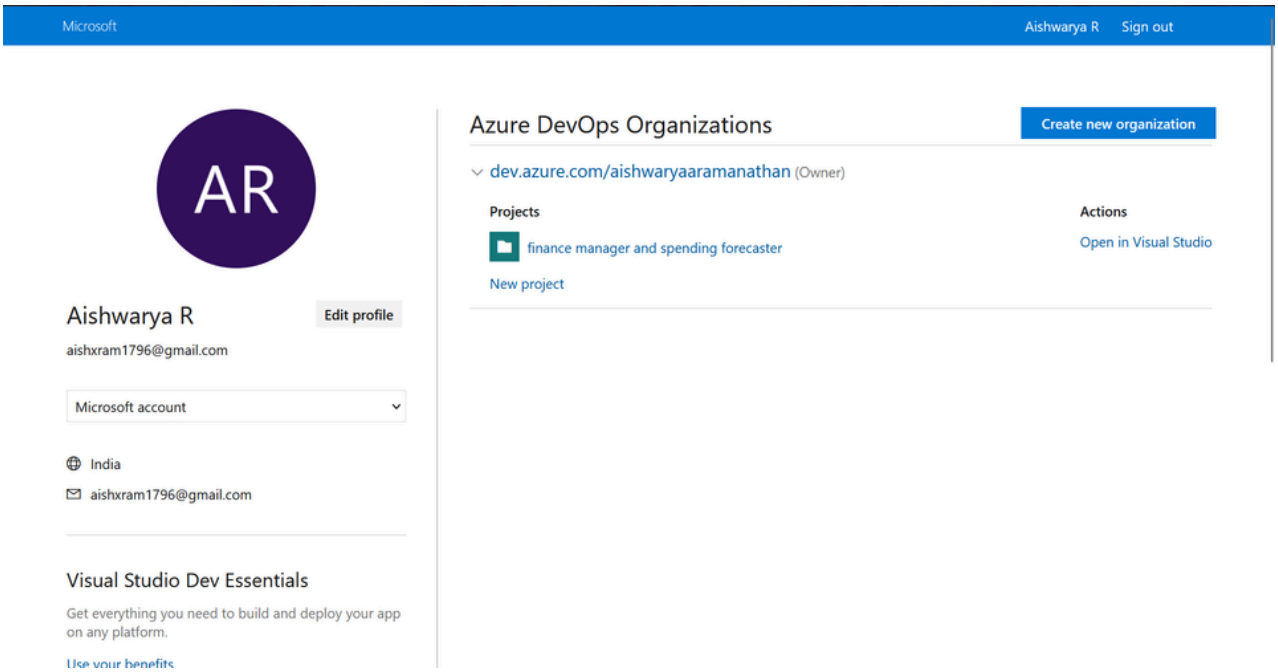
EXPNO: 1	AZUREDEVOPS ENVIRONMENT SETUP

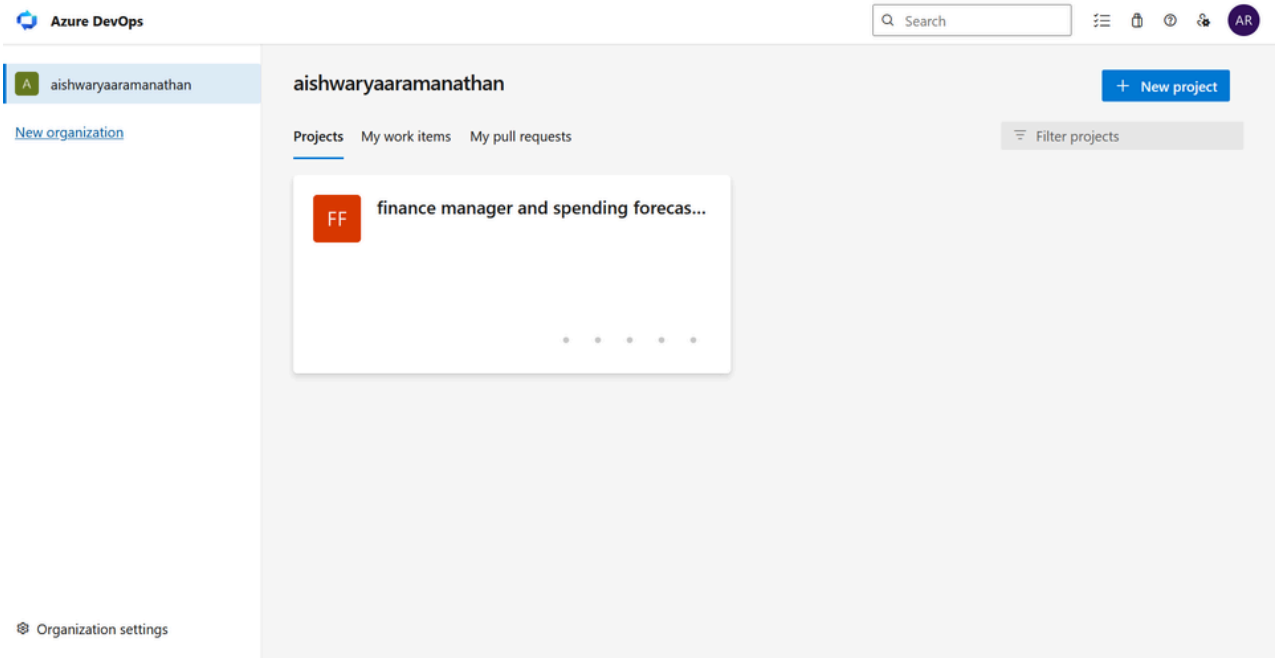
Aim:

To set up and access the Azure DevOps environment by creating an organization through the Azure Portal.

Procedure:







Result:

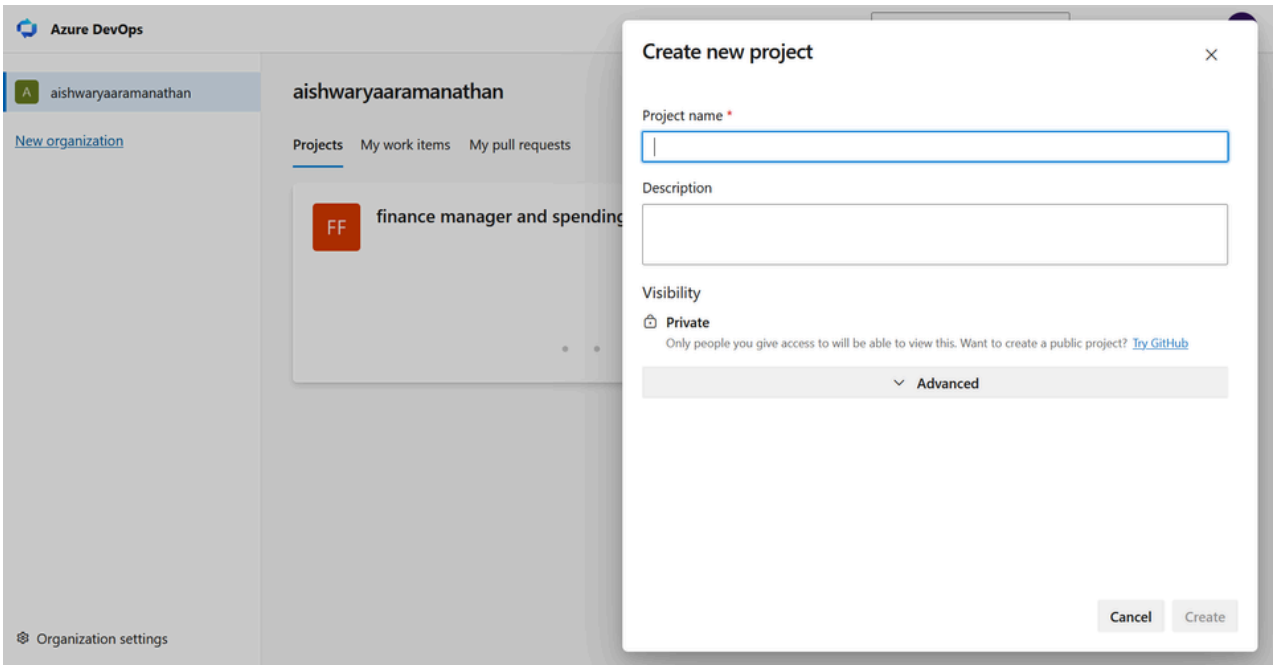
Successfully accessed the Azure DevOps environment and created a new organization through the Azure Portal.

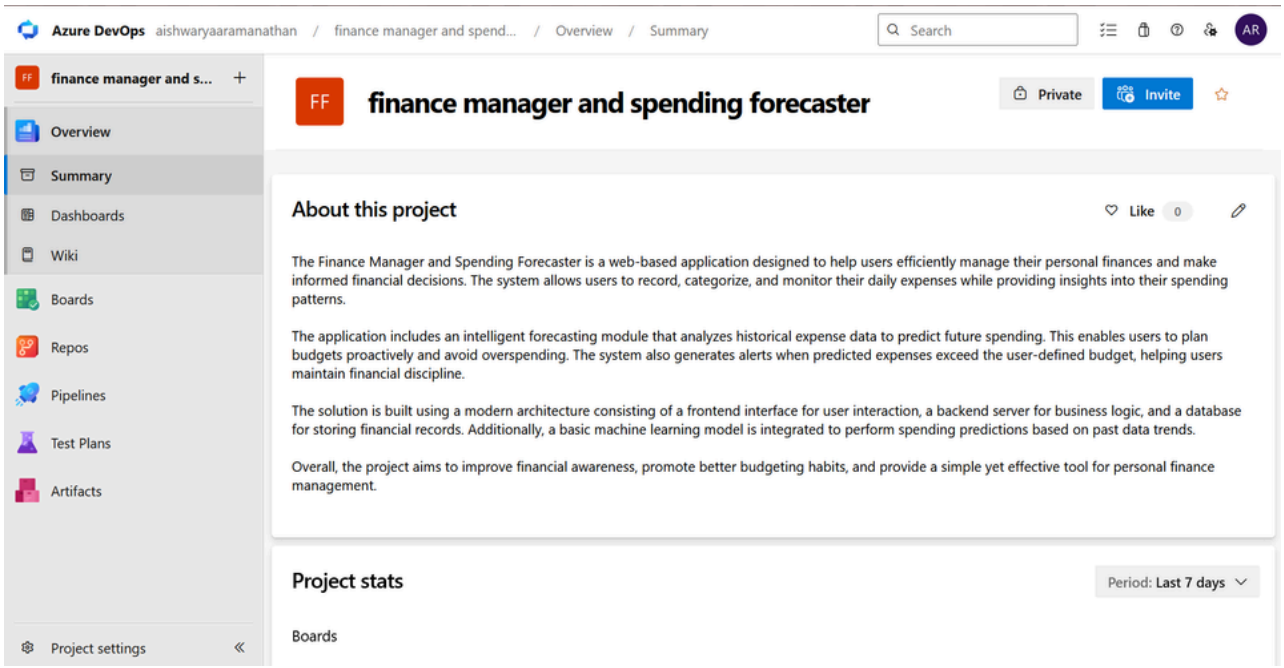
EXPNO: 2	AZUREDEVOPS PROJECT SETUP AND USER STORY MANAGEMENT

Aim:

To set up an Azure DevOps project for efficient collaboration and agile work management.

Procedure:





Result:

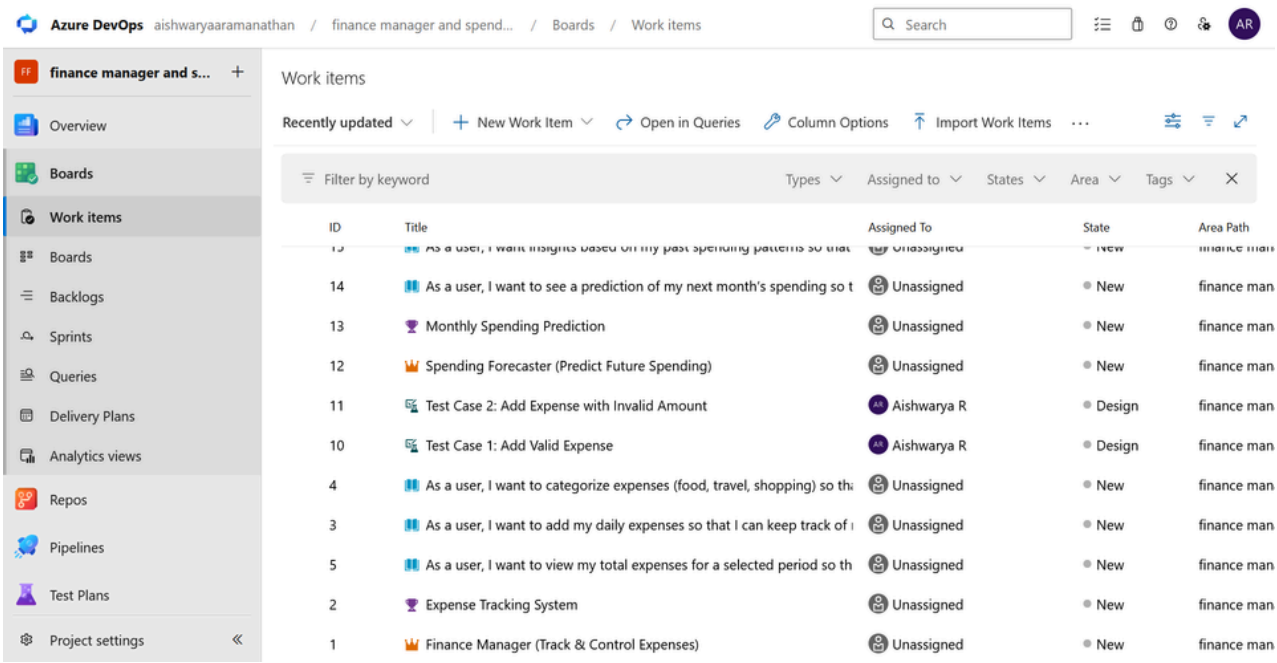
Successfully created an Azure DevOps project with user story management and agile workflow setup

EXPNO: 3	SETTINGUPEPICS, FEATURES, AND USER STORIES FOR PROJECT PLANNING

Aim:

To learn about how to create epics, user story, features, backlogs for your assigned project.

Procedure:



Azure DevOps aishwaryaaramanathan / finance manager and spend... / Boards / Work items

Search

finance manager and s...

Overview
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EPIC 1

1 Finance Manager (Track & Control Expenses)

No one selected | 0 Comments | Add Tag | Save | Follow

Updated by Aishwarya R: 2h ago

State: New
Reason: New
Area: finance manager and spending forecaster
Iteration: finance manager and spending forecaster

Details | 1 | 0

switch to Markdown editor

Start Date: Select a date...
Target Date: Select a date...

Classification

Value area: Business

Development

Add link

Link an Azure Repos commit, pull request or branch to see the status of your development. You can also create a branch to get started.

Related Work

Add link

Add an existing work item as a parent

Child

2 Expense Tracking System

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Search

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Boards
Backlogs
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Test Plans
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Recently updated | Back to Work Items | 10 of 18

FEATURE 13

13 Monthly Spending Prediction

No one selected | 0 Comments | Add Tag | Save | Follow

Updated by Aishwarya R: 2h ago

State: New
Reason: New
Area: finance manager and spending forecaster
Iteration: finance manager and spending forecaster

Details | 4 | 0

Description

Click to add Description.

Discussion

Add a comment. Use # to link a work item, @ to mention a person, or ! to link a pull request.

switch to Markdown editor

Planning

Priority: 2
Risk
Effort
Business Value
Time Criticality
Start Date: Select a date...
Target Date

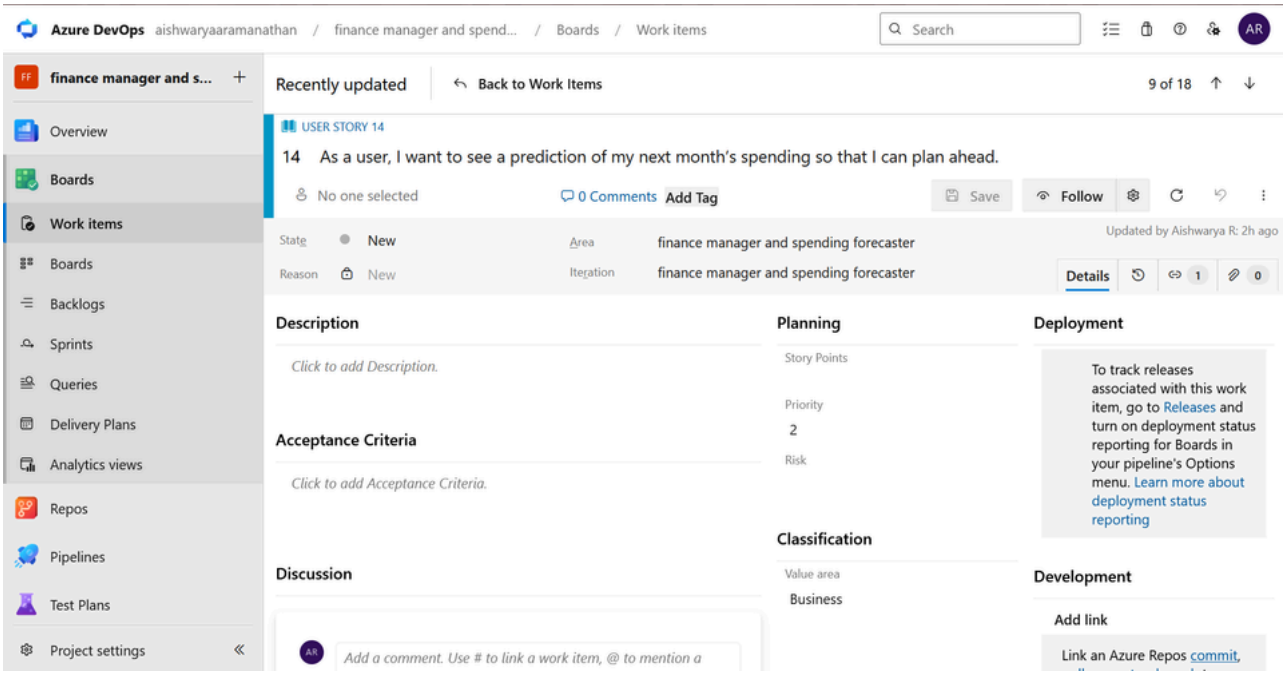
Deployment

To track releases associated with this work item, go to Releases and turn on deployment status reporting for Boards in your pipeline's Options menu. Learn more about deployment status reporting

Development

Add link

Link an Azure Repos commit,



Result:

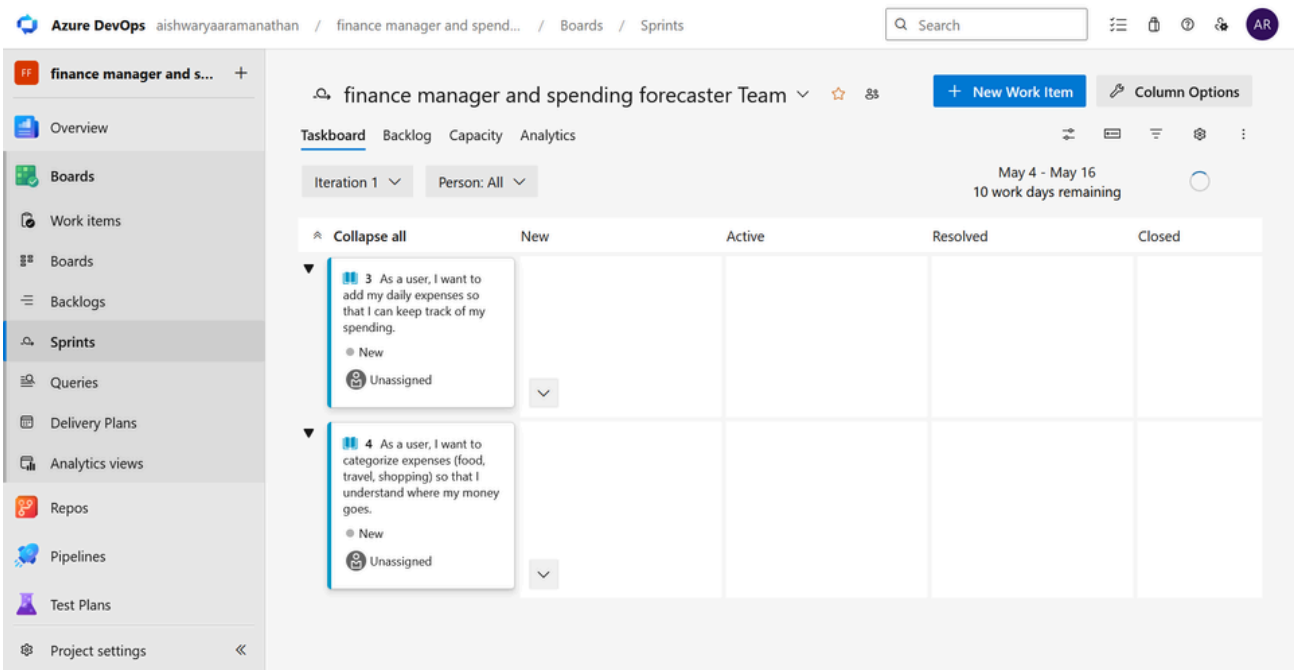
Thus, the creation of epics, features, user story and task has been created successfully.

EXPNO: 4	SPRINT PLANNING

Aim:

To assign user story to specific sprint for finance manager and spending tracker Project.

Procedure:



Result:

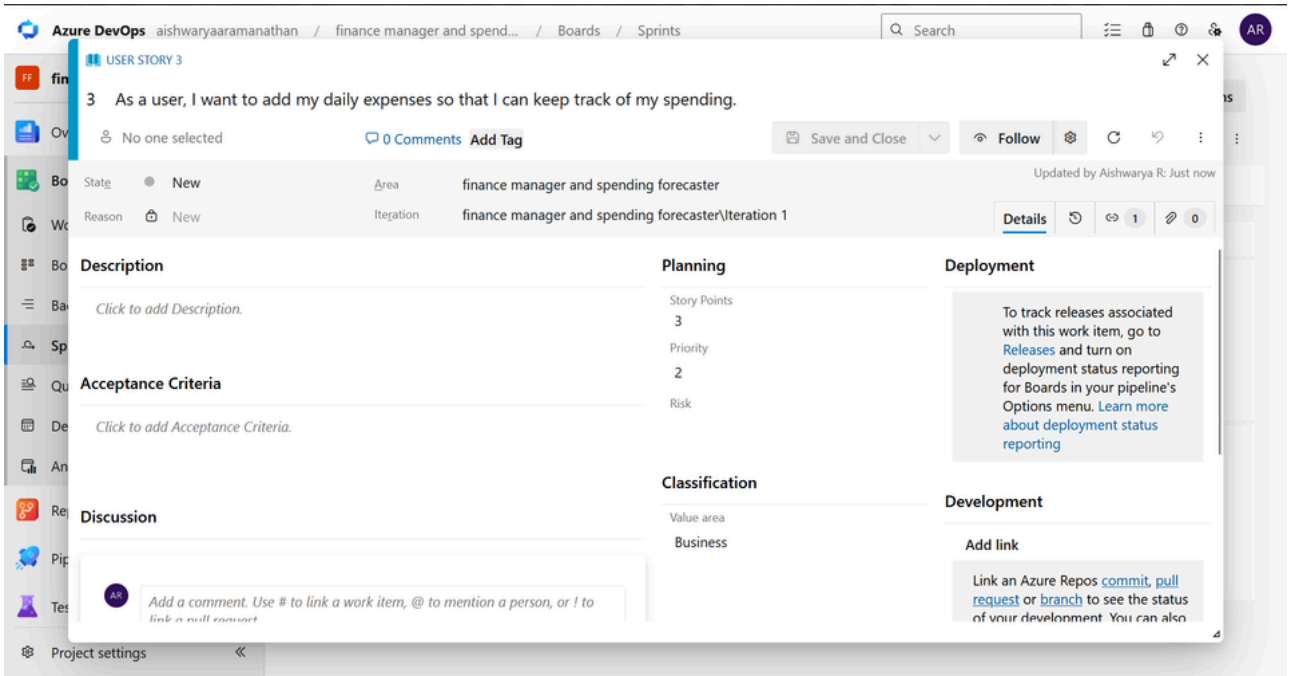
The Sprints are created for Finance manager and spending tracker Project.

EXPNO: 5	POKER ESTIMATION

Aim:

Create Poker Estimation for the user stories - Finance Manager and Spending Tracker Project.

Procedure:



Result:

The Estimation/Story Points is created for the project using Poker Estimation.

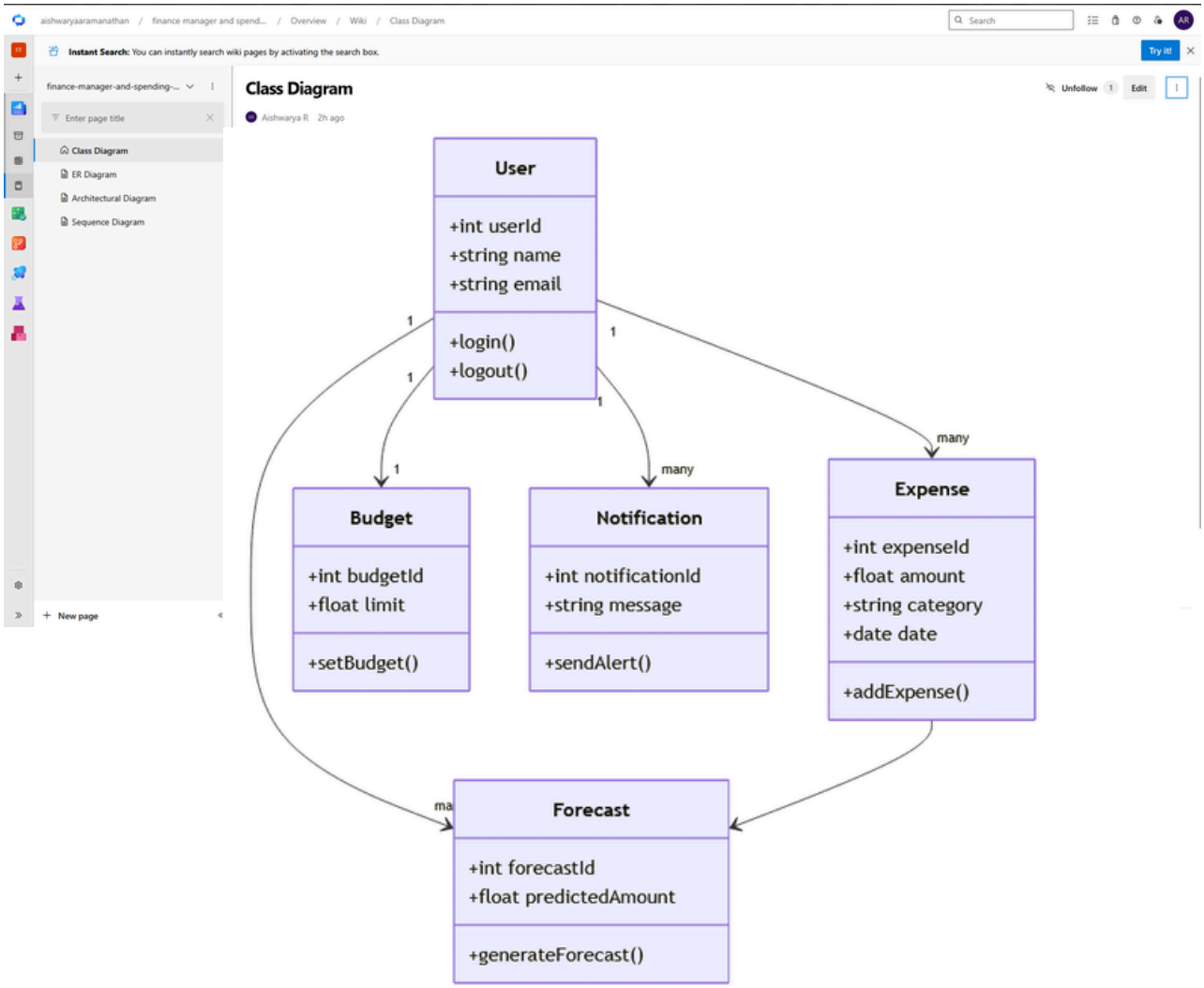
EXPNO: 6	DESIGNINGCLASS AND SEQUENCE DIAGRAMS FOR PROJECT ARCHITECTURE

Aim:

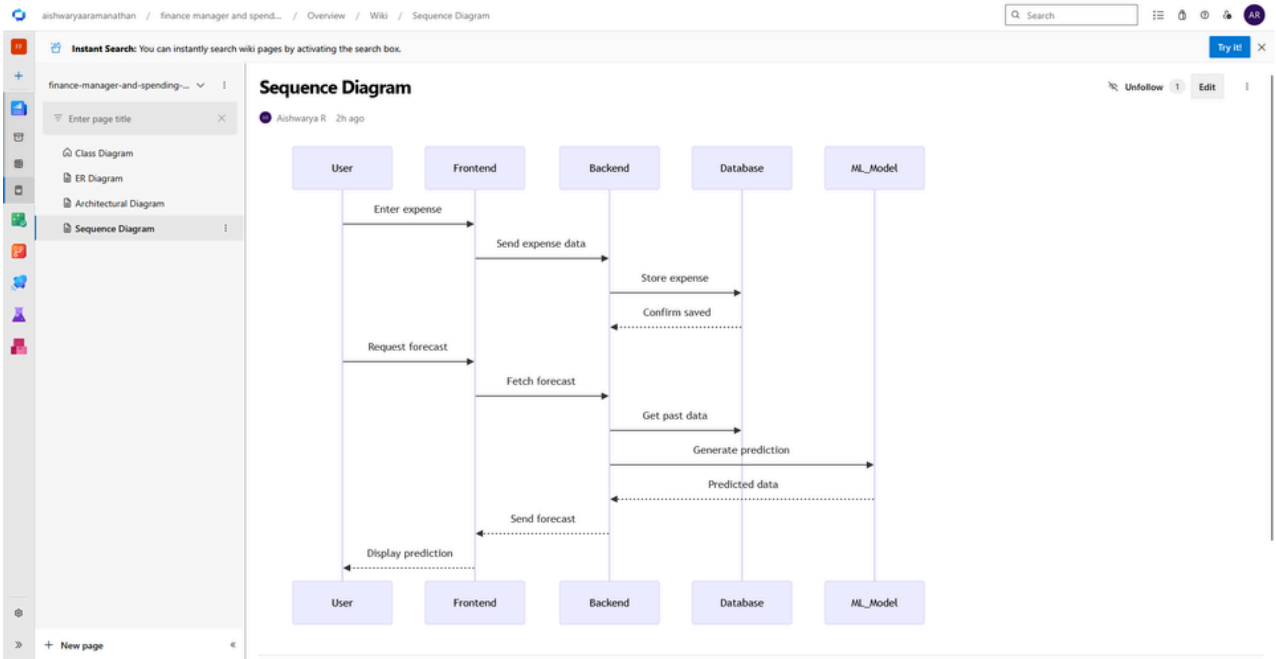
To Design a Class Diagram and Sequence Diagram for the given Project.

Procedure:

6A. Class Diagram



6B. Sequence Diagram



Result:

The Class Diagram and Sequence Diagram is designed Successfully for the Project.

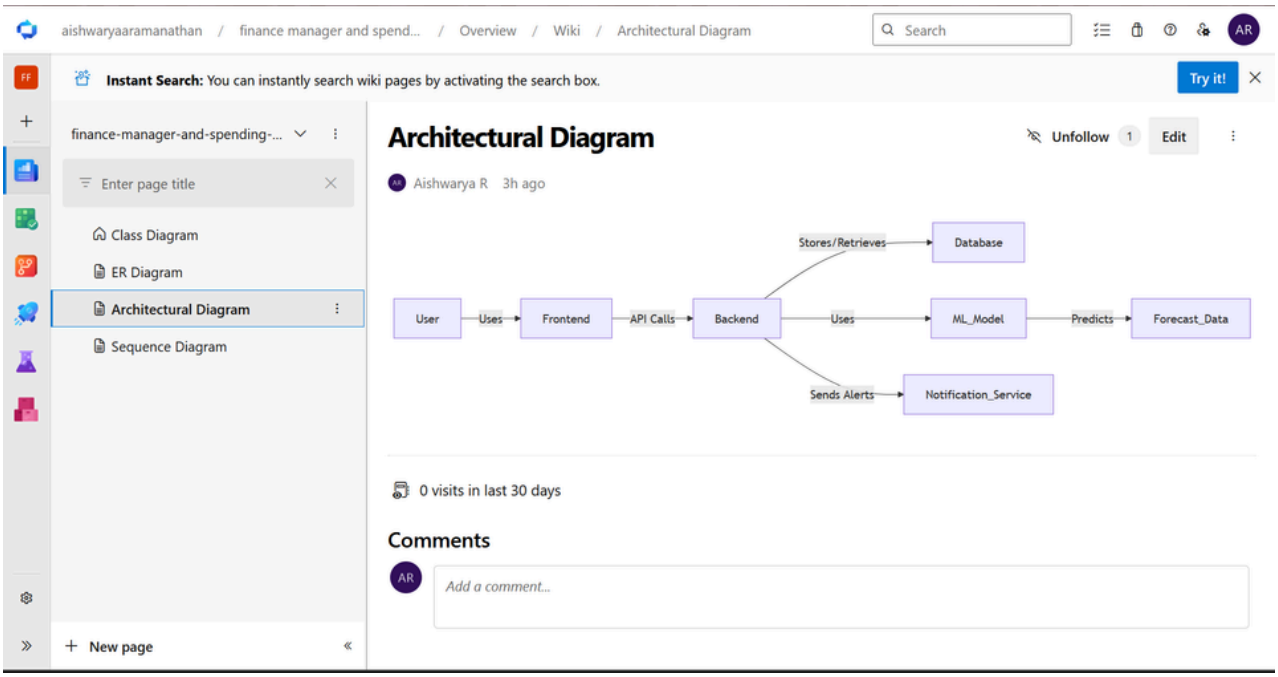
EXPNO: 7	DESIGNINGARCHITECTURAL AND ER DIAGRAMS FOR PROJECT STRUCTURE

Aim:

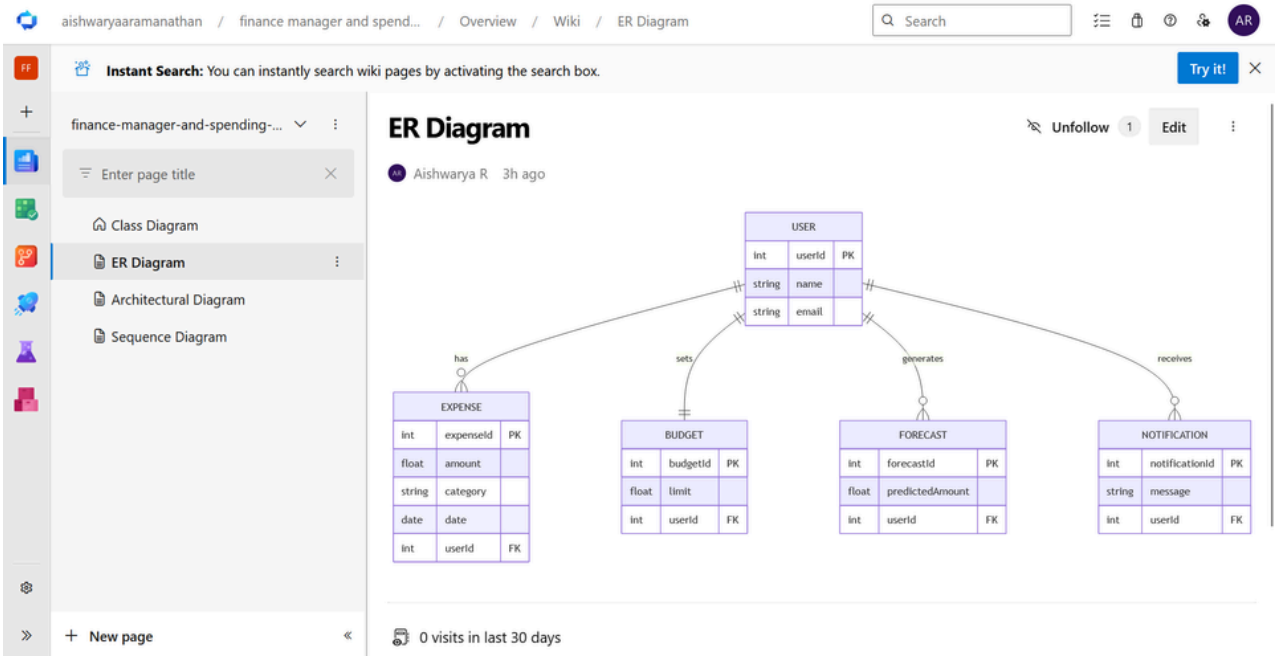
To Design an Architectural Diagram and ER Diagram for the given Project.

Procedure:

7A. Architectural Diagram



7B.ER Diagram



Result:

The Architecture Diagram and ER Diagram is designed Successfully for the Project.

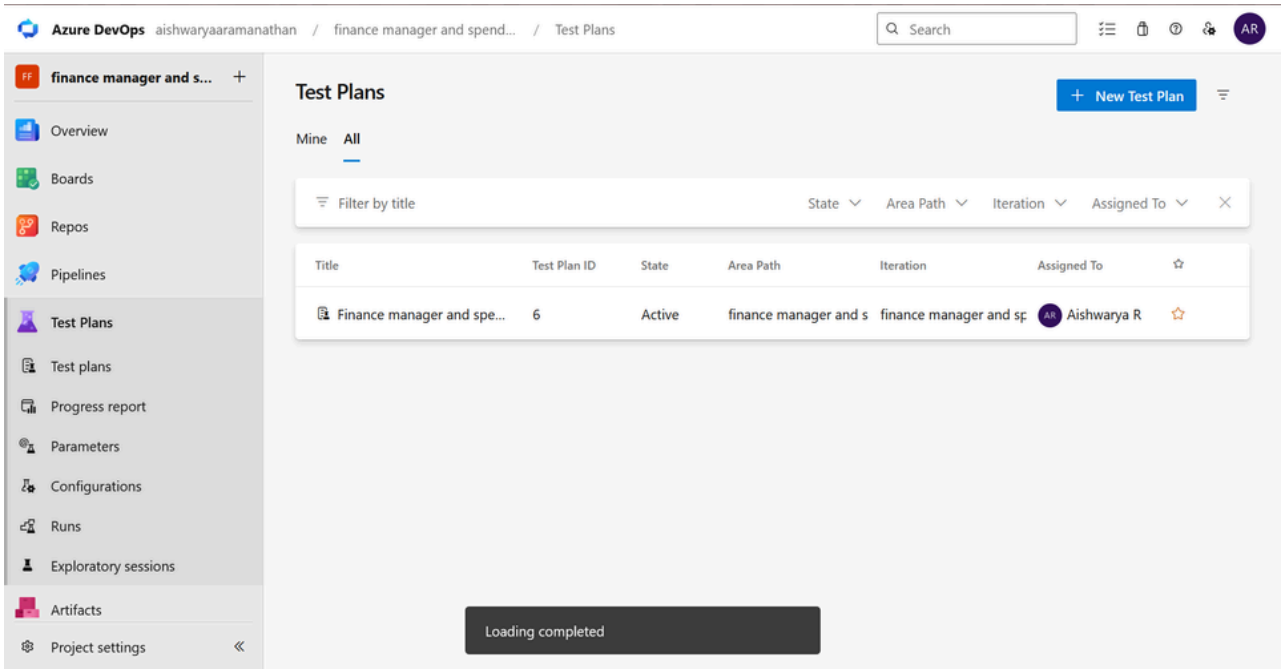
EXPNO: 8	TESTING – TEST PLANS AND TEST CASES

Aim:

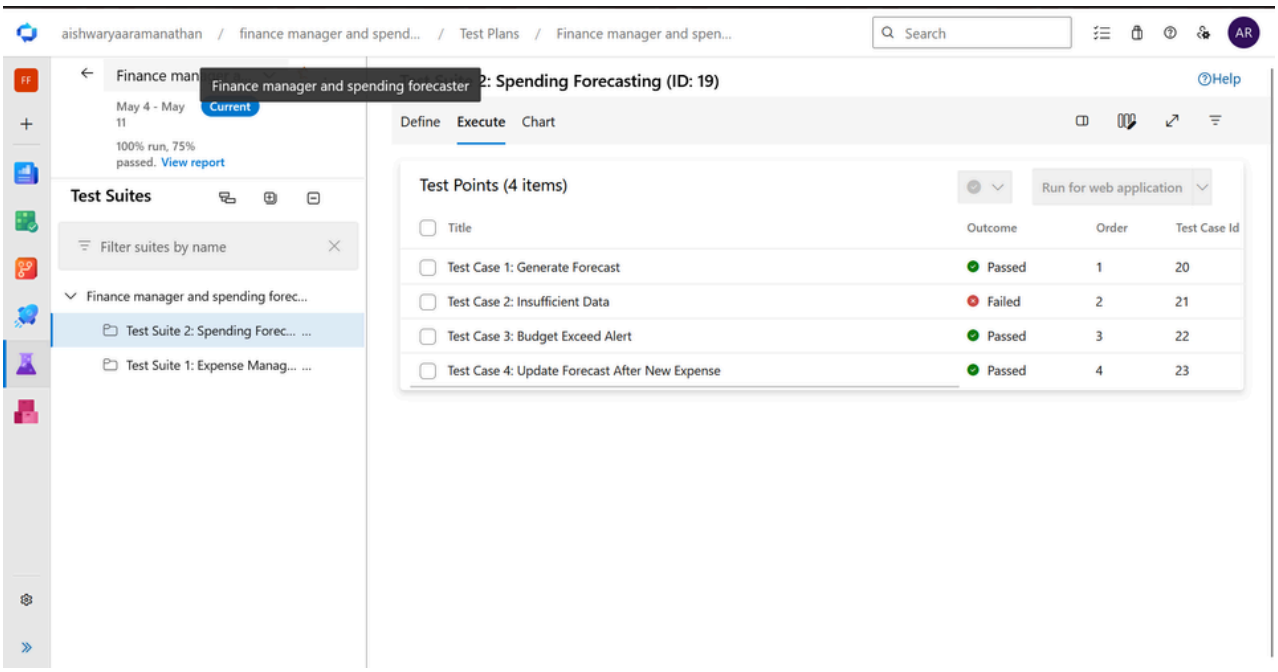
Test Plans and Test Case and write two test cases for at least five user stories showcasing the happy path and error scenarios in azure DevOps platform.

Procedure:

1.New test plan



2.Test suite



3.Test case

Test Suite 1: Expense Management
Objective: Verify that users can correctly add, categorize, and view expenses.

Test Case 1: Add Valid Expense

- Test ID: TC_EM_01
- Description: Add a valid expense entry
- Input: Amount = 500, Category = Food, Date = Today
- Expected Result: Expense is successfully stored and displayed

Test Case 2: Add Expense with Invalid Amount

- Test ID: TC_EM_02
- Description: Enter negative amount
- Input: Amount = -200
- Expected Result: Error message “Invalid amount” is shown

Test Case 3: Missing Category

- Test ID: TC_EM_03
- Description: Submit expense without category
- Input: Amount = 300, Category = null
- Expected Result: System prompts user to select a category

Test Case 4: View Expense Summary

- Test ID: TC_EM_04
- Description: View total expenses for a period
- Input: Date range = Last 7 days
- Expected Result: Correct total expense is displayed

Test Suite 2: Spending Forecasting

Objective: Ensure prediction logic and alerts work correctly.

Test Case 1: Generate Forecast

- Test ID: TC_SF_01
- Description: Predict next month spending
- Input: Past 3 months expense data
- Expected Result: Forecast value is generated and displayed

Test Case 2: Insufficient Data

- Test ID: TC_SF_02
- Description: Forecast with minimal data
- Input: Only 1 expense record
- Expected Result: Warning "Not enough data for prediction"

Test Case 3: Budget Exceed Alert

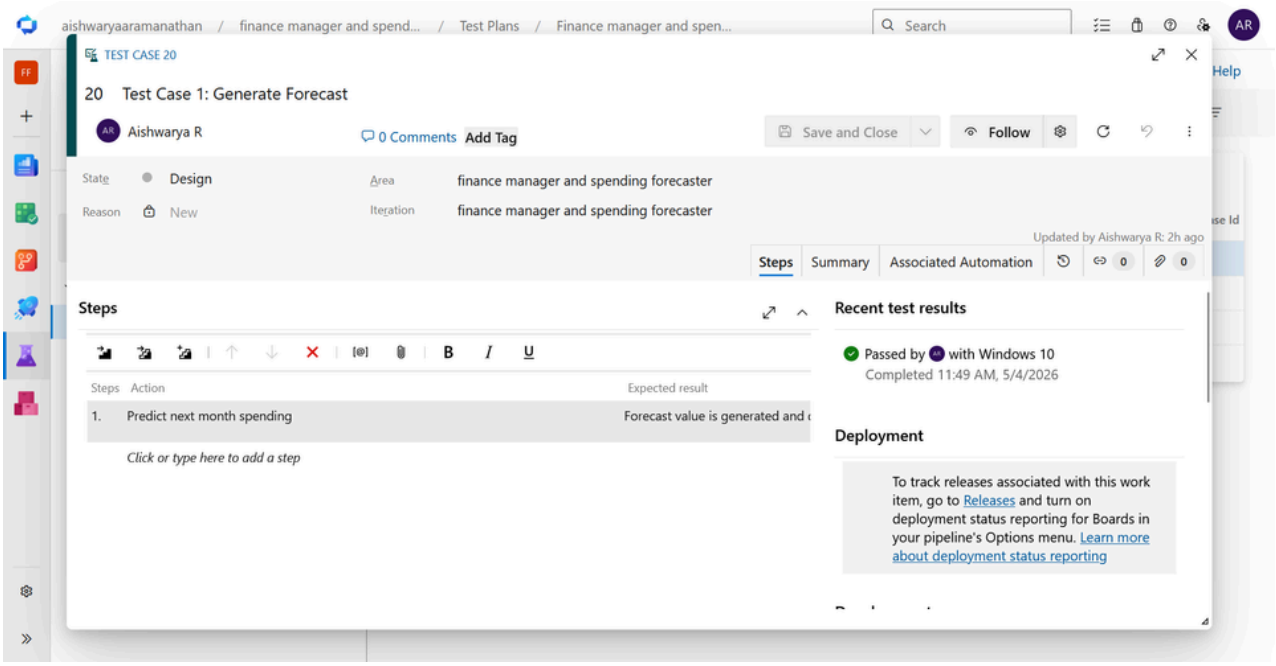
- Test ID: TC_SF_03
- Description: Predicted spending exceeds budget
- Input: Budget = 5000, Prediction = 7000
- Expected Result: Alert notification is triggered

Test Case 4: Update Forecast After New Expense

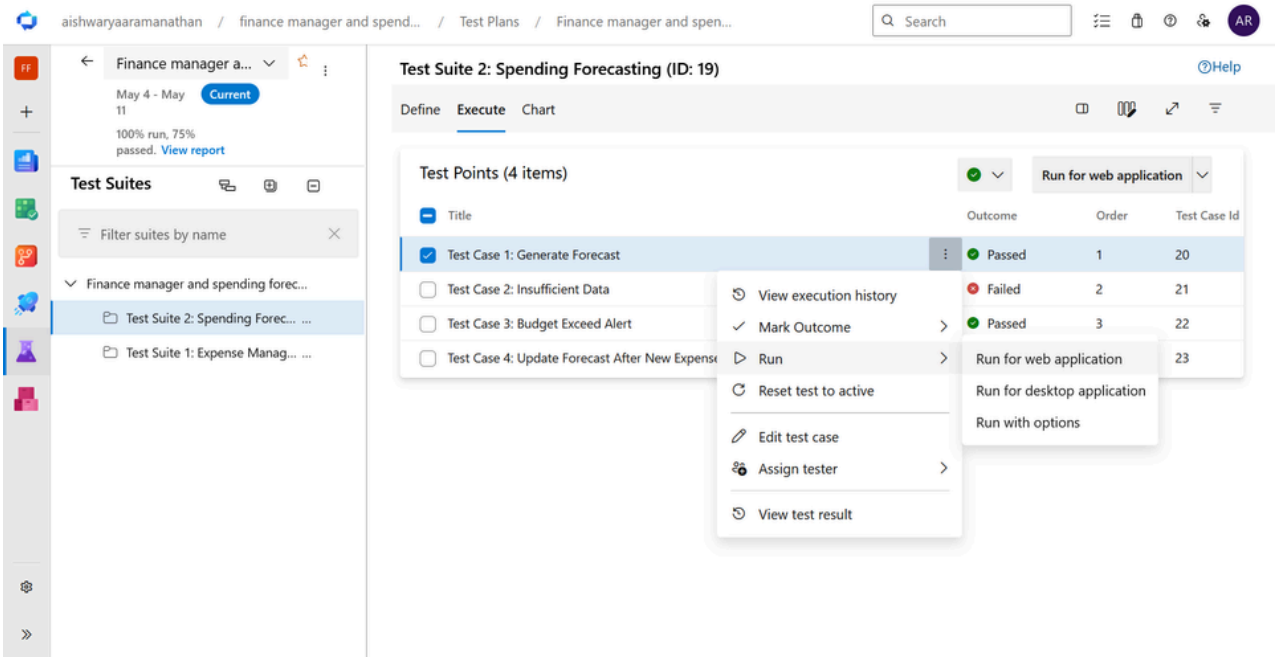
- Test ID: TC_SF_04
- Description: Add new expense and regenerate forecast
- Input: Add expense = 1000
- Expected Result: Forecast updates accordingly

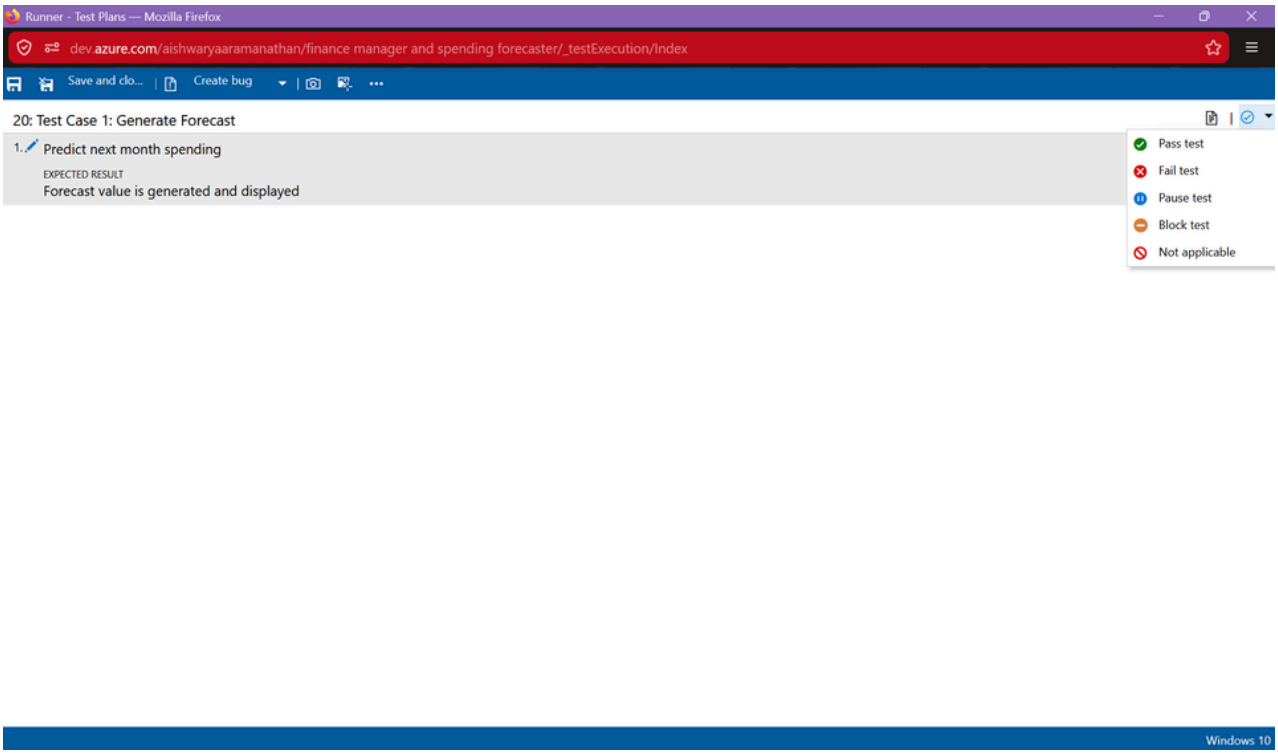
Result:

The Architecture Diagram and ER Diagram is designed Successfully for the Project.

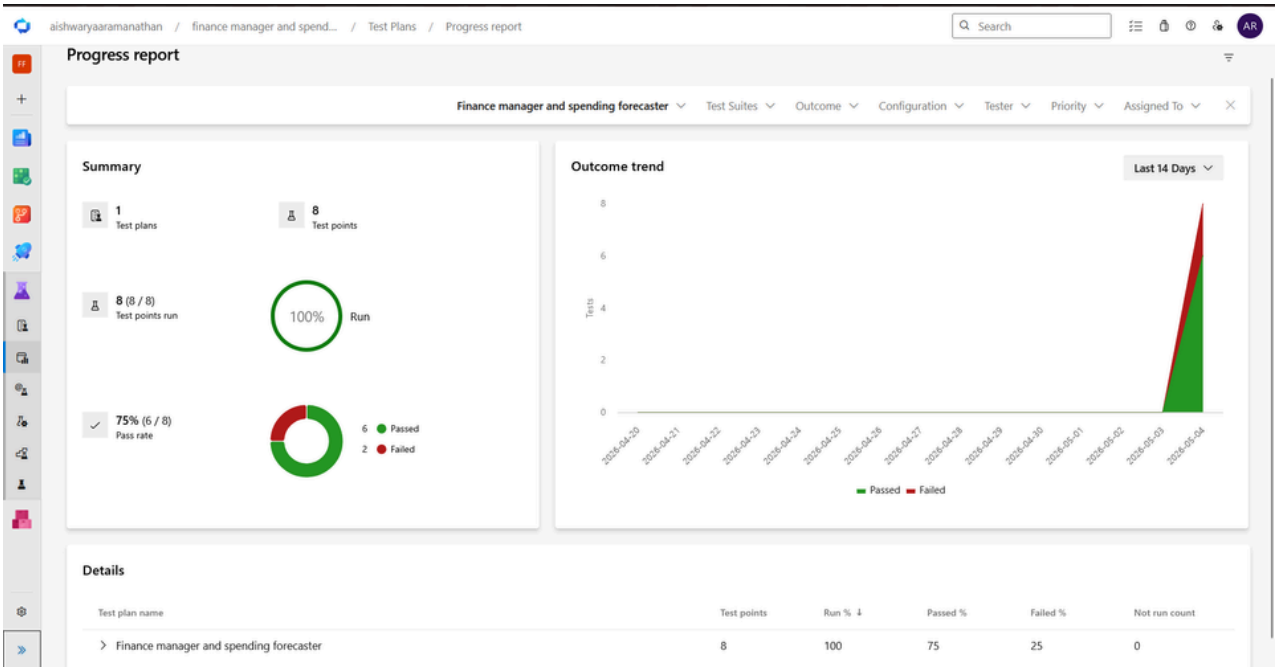


4.Run Test case





4.Progress Report



Result:

The test plans and test cases for the user stories is created in Azure DevOps with Happy Path and error path.

EXPNO:9	LOADTESTING AND PERFORMANCE TESTING

Aim:

To create an Azure Load Testing resource and run a load test to evaluate the performance of a target endpoint

Procedure:

Result:

Successfully created the Azure Load Testing resource and executed a load test to assess the performance of the specified endpoint.

EXPNO: 10	GITHUB:PROJECT STRUCTURE & NAMING CONVENTIONS

Aim:

To provide a clear and organized view of the project's folder structure and file naming conventions, helping contributors and users easily understand, navigate, and extend the project.

Procedure:

GitHub Project Structure

Result:

The GitHub repository clearly displays the organized project structure and consistent naming conventions, making it easy for users and contributors to understand and navigate the codebase.